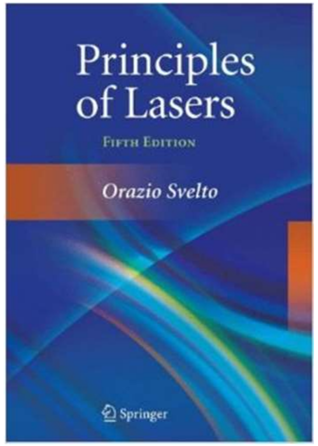
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers



Ch2-2
Open Cavity & Gaussian Beam

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Chapter 2: Open Cavity & Gaussian Beam

❖ 2.3 Mode for Open Cavity & Diffraction Theory

- Introduction
- The self-reconstruction of mode
- Fresnel Kirchhoff's law
- The physical meaning of the integral equation

❖ 2.5 The self-reconstructed mode of square confocal cavity

- Variables separation
- Solutions: Analytical solution and approximation solution

1. Introduction

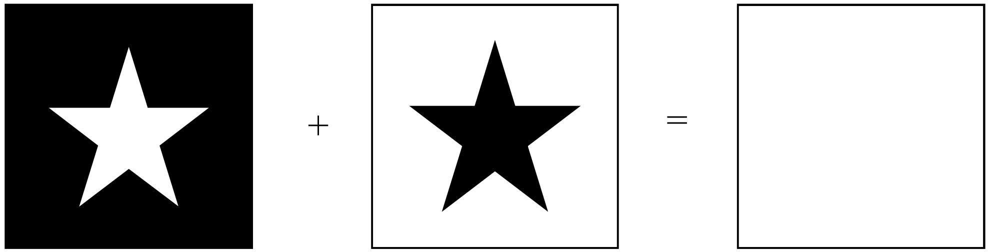
Diffraction plays an important role in spatial distribution of laser energy

After multiple traveling, a steady-state field will be formed: energy distribution will not be influenced by diffraction any more; the field of light is the same as the field before previous reflection

- The shape of the field is the same
- The amplitude is different
- The phase is difference

Babinet's Principle: In physics, Babinet's principle is a theorem concerning diffraction. It mentions that the diffraction pattern from an opaque body is identical to that from a hole of the same size and shape except for the overall forward beam intensity.

Complementary (互补)

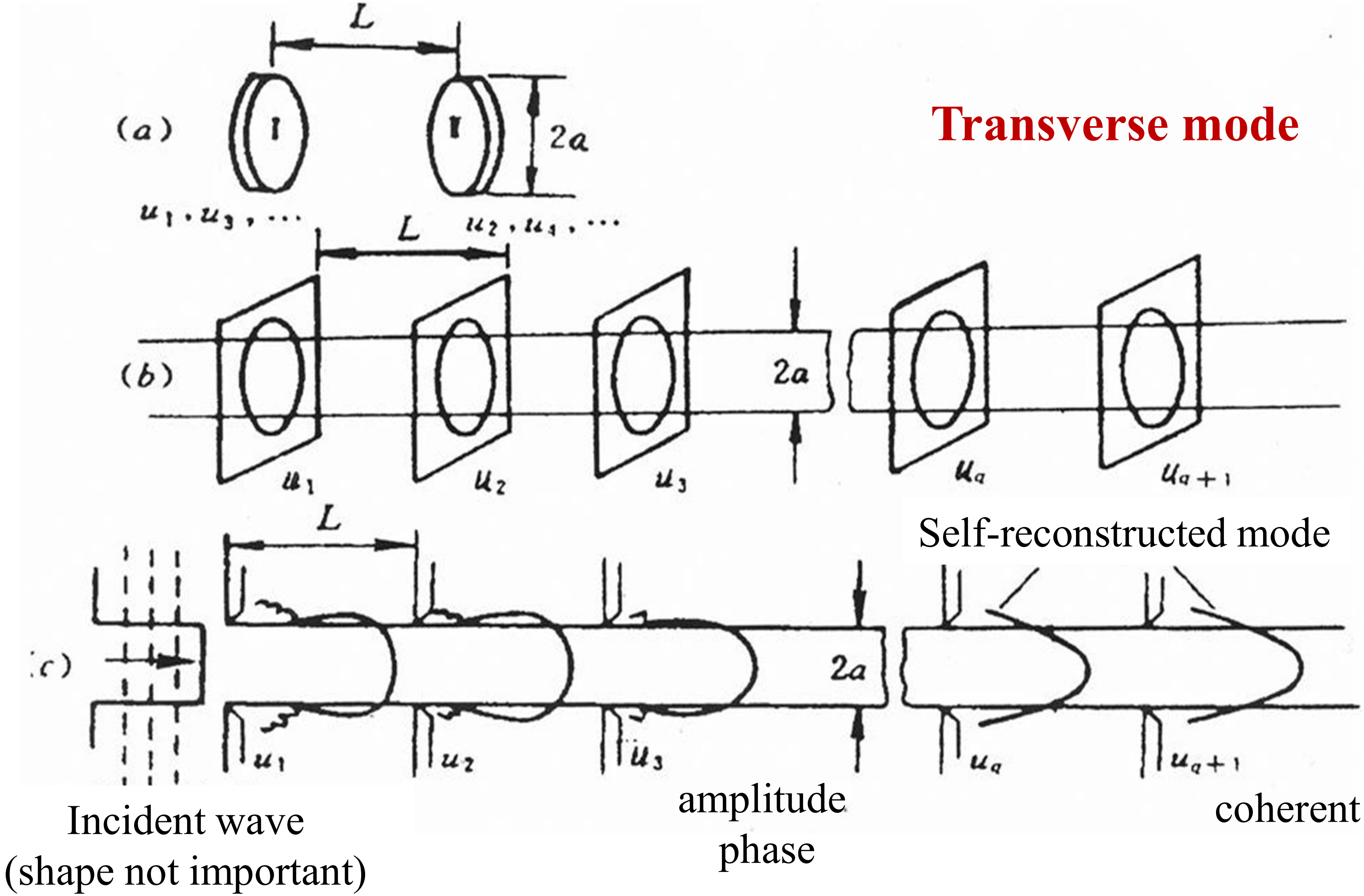


$$-\frac{\mathrm{i}}{\lambda}\iint_{\Sigma_A} U_0(Q)\frac{\mathrm{e}^{ikr}}{r}\mathrm{d}\Sigma-\frac{\mathrm{i}}{\lambda}\iint_{\Sigma_B} U_0(Q)\frac{\mathrm{e}^{ikr}}{r}\mathrm{d}\Sigma=-\frac{\mathrm{i}}{\lambda}\iint_{\Sigma_0} U_0(Q)\frac{\mathrm{e}^{ikr}}{r}\mathrm{d}\Sigma$$

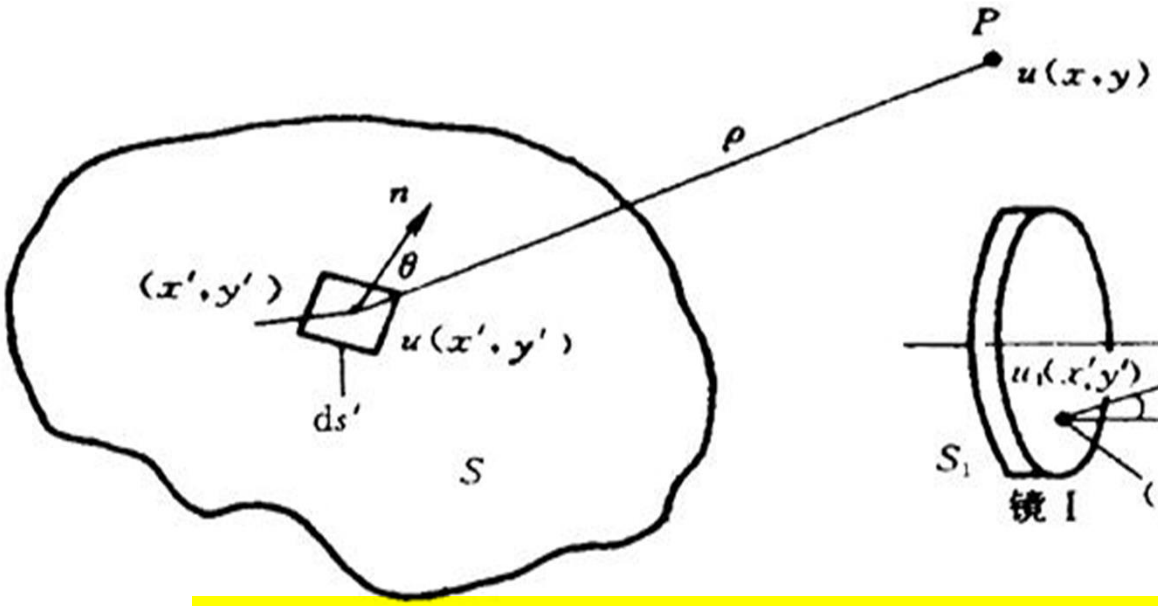
$$U_A(P)+U_B(P)=U_0(P)$$

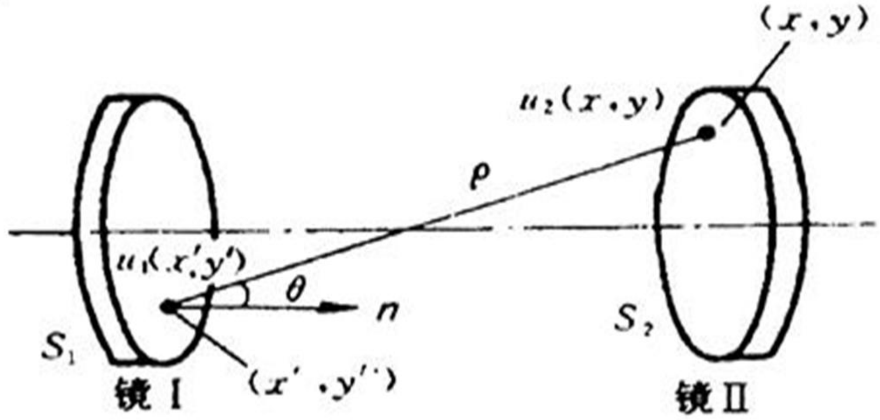
The far-field diffraction patterns induced by a pair of complementary structures have the same shape

2. The Self-Reconstruction of Mode



3. Fresnel Kirchhoff's Law





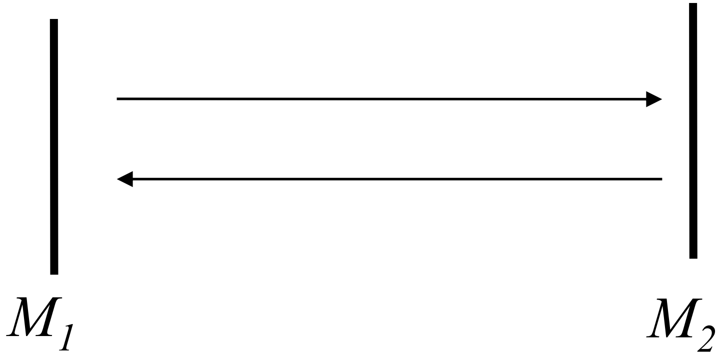
$$u(x,y)=\frac{ik}{4\pi}\iint_{s}\underline{u(x',y')}\frac{e^{-ik\rho}}{\rho}\underline{(1+\cos\theta)}ds'$$

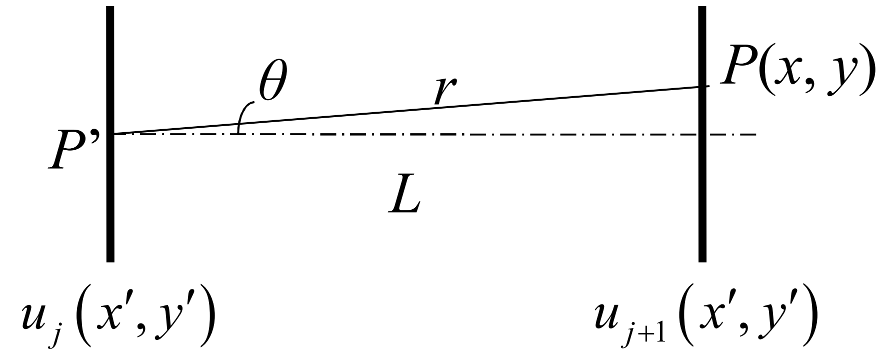
Light field distribution function on the surface S

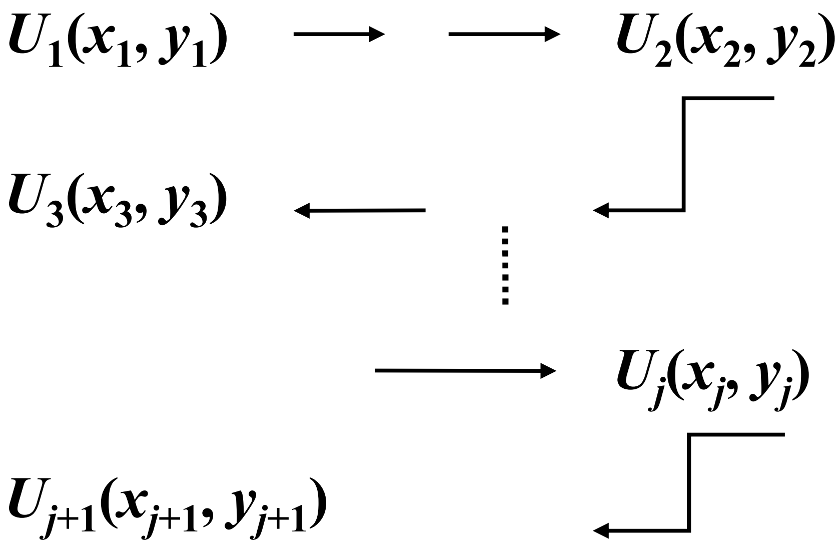
Spherical wave from each point source

Angle-related factor

The idea of self-reconstruction of mode







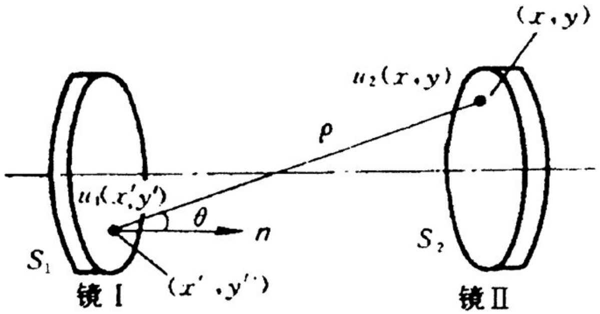
<u>The same light field</u>

The self-reconstruction of mode

With multiple travels, beside the decrease of amplitude and change of phase, the light field U_{j+1} can construct U_j

Condition for reconstructed mode

$$\boxed{U_{j+1}(x, y) = \frac{1}{\gamma} U_j(x, y)}$$



- Fresnel Kirchhoff's law
- self-reconstruction of mode

⇨ The integral equation of self-reconstruction field in optical cavity

$$u_{j+1}(x,y)=\frac{ik}{4\pi}\iint_s u_j(x',y')\frac{e^{-ik\rho}}{\rho}(1+\cos\theta)\,ds'$$

The integral equation

$$u_j(x,y)=\gamma\frac{ik}{4\pi}\iint_s u_j(x',y')\frac{e^{-ik\rho}}{\rho}(1+\cos\theta)\,ds'$$

$$u_{j+1}(x,y)=\gamma\frac{ik}{4\pi}\iint_s u_{j+1}(x',y')\frac{e^{-ik\rho}}{\rho}(1+\cos\theta)\,ds'$$

…….. ……..

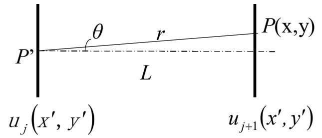
The integral of the self-reconstructed mode

$$V(x,y)=\gamma\frac{ik}{4\pi}\iint V(x',y')\frac{e^{-ik\rho}}{\rho}(1+\cos\theta)\,ds'$$

The stable field distribution in cavity

The integral of the self-reconstructed mode

$$V(x,y)=\gamma\frac{ik}{4\pi}\iint V(x',y')\frac{(1+\cos\theta)}{\rho}e^{-ik\rho}ds'$$



$$\because L>>x,y \qquad \rho\approx L,\cos\theta\approx 1 \quad\Rightarrow\quad (1+\cos\theta)/\rho\approx 2/L$$

$$V(x,y)=\gamma\frac{i}{\lambda L}\iint V(x',y')e^{-ik\rho(x,y,x',y')}ds'$$

$$\boxed{V(x,y)=\gamma\iint K(x,y,x',y')V(x',y')ds'}$$

where $K(x,y,x',y')=\dfrac{i}{\lambda L}e^{-ik\rho(x,y,x',y')}$ Kernel (核) of the integral equation

- Capable for any symmetric cavity (Parallel plane, confocal, general spherical mirror cavity)
- Find $V(x,y)$ $\Leftrightarrow$ mode distribution $\Rightarrow$ Solve the Integral Equation

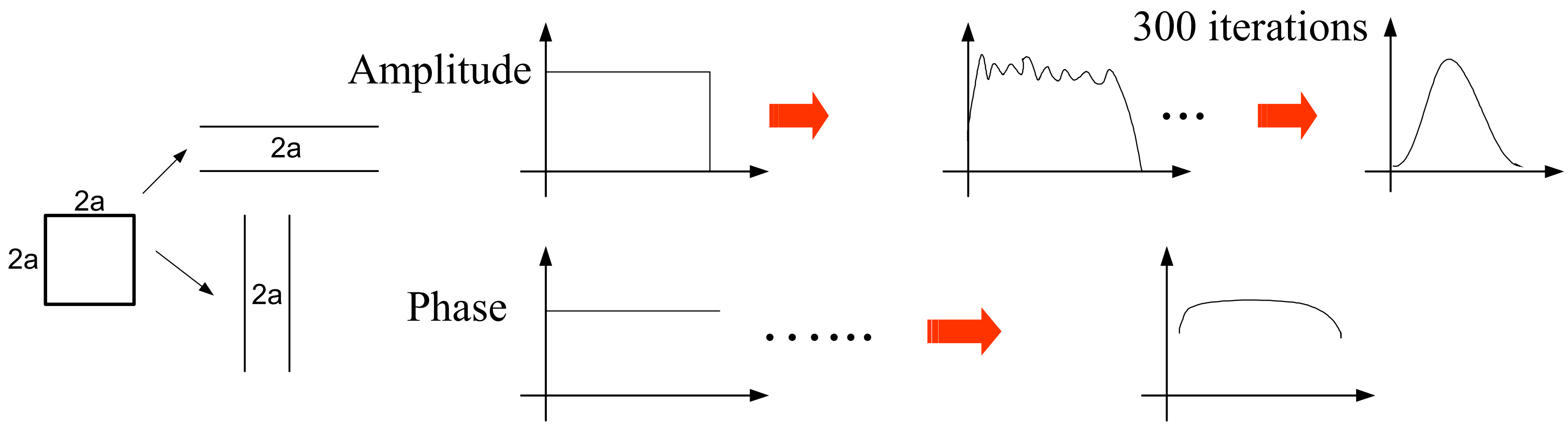
Solve the integral of the self-reconstructed mode?

1. The analytical solution: only for symmetrical confocal cavity

Type of Confocal Cavity	Analytical Solution	Approximation
Square	Prolate-ellipsoid function	Hermite-Gaussian function
Circular	Hyper-ellipsoid function	Laguerre-Gaussian function

The solution for other stable spherical mirror cavities can be obtained by means of "equivalent" to the symmetry confocal cavity.

2. The numerical solution: numerical iteration method (迭代)



4. Integral equation: physical meaning

$$V(x,y)=\gamma\frac{i}{\lambda L}\iint V(x',y')e^{-ik\rho(x,y,x',y')}ds'$$

Eigen Function
本征函数

Eigen Value
本征值

1) The expression of Eigen function

$$V_{mn}(x,y)=A_{mn}(x,y)e^{i\varphi_{mn}(x,y)}$$

$V_{mn}(x, y)$ Field distribution on the mirror (Eigen function ⟺ transverse mode)

$A_{mn}(x, y)$ Amplitude distribution on the mirror

$\varphi_{mn}(x, y)$ Phase distribution on the mirror

- There is no analytical solution for general spherical mirror cavity
- Analytical solution can be obtained only for symmetrical confocal cavity

2) The meaning of eigen value γ — complex constant

$$\gamma_{mn} = \left|\gamma_{mn}\right| e^{i\delta\varphi_{mn}}$$ — Transmission factor γ is related to δ_D

$$if \quad \gamma_{mn} = e^{\alpha + i\beta} \qquad u_{j+1} = \frac{1}{\gamma_{mn}} u_j \Rightarrow u_{j+1} = \underbrace{\left(e^{-\alpha} u_j\right)}_{\text{decrease of amplitude per round}} \underbrace{e^{-i\beta}}_{\text{phase shift}}$$

$$\boxed{\delta_D \equiv \frac{\left|u_{mnj}\right|^2 - \left|u_{mnj+1}\right|^2}{\left|u_{mnj}\right|^2} = 1 - e^{-2\alpha} = 1 - \frac{1}{\left|\gamma_{mn}\right|^2}}$$

δ_D — The relative power loss per travel—单程损耗

$\left|\gamma_{mn}\right|$ — The single-pass loss of self-reconstructed mode

— Different transverse mode has different δ_D

— $\left|\gamma_{mn}\right| \uparrow \rightarrow$ 模的单程损耗 $\delta_D \uparrow$

3) The phase shift $\delta\varphi_{mn}$ — the total phase shift of the self reconstructed modes

$$\delta\varphi_{mn} = \arg u_{j+1} - \arg u_j$$

$$\because \gamma = e^{\alpha + i\beta} \qquad u_{j+1} = \frac{1}{\gamma} u_j$$

$$\delta\varphi_{mn} = -\beta = \arg \frac{1}{\gamma}$$

Geometrical phase shift
几何相移

Additional phase shift
附加相移

$$\delta\varphi_{mn} = -\beta = -kL + \Delta\varphi = -\frac{2\pi L}{\lambda} + \underline{\Delta\varphi}$$

Resonant condition $\delta\varphi_{mn} = q\pi$

If we know γ_{mn}, can obtain the resonant frequency of the reconstructed mode

The modulus of the complex constant is the one-round loss of the self reconstructed mode, and its argument measures the phase shift, which also determines the resonant frequency of the mode.

Variable separation

$$V(x,y)=\gamma\iint K(x,y,x',y')V(x',y')\,ds'$$

When $\quad L,R>>a \qquad K(x,y,x',y')=\dfrac{i}{\lambda L}e^{-ik\rho(x,y,x',y')}$

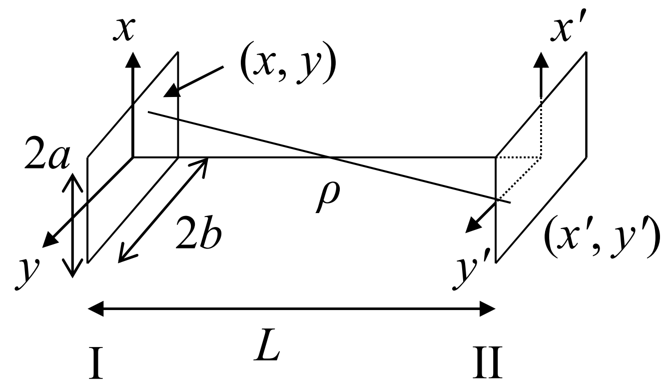
$if \quad \rho(x,y,x',y')\overset{?}{=}\rho_x(x,x')\rho_y(y,y')$

$$K(x,y,x',y')=\frac{i}{\lambda L}e^{-ik\rho(x,y,x',y')} \Rightarrow K(x,y,x',y')=K_x(x,x')K_y(y,y')$$

$assume$

$$V(x,y)=V(x)V(y) \qquad \begin{cases} V(x)=\gamma_x\int_{-a}^{+a}K_x(x,x')V(x')\,dx' \\ V(y)=\gamma_y\int_{-b}^{+b}K_y(y,y')V(y')\,dy' \end{cases}$$

1) Rectangular Planar Cavity



$$\rho = \sqrt{\left(x - x'\right)^2 + \left(y - y'\right)^2 + L^2}$$

$$\rho = \sqrt{\left(x - x'\right)^2 + \left(y - y'\right)^2 + L^2} = L\sqrt{\left(\frac{x - x'}{L}\right)^2 + \left(\frac{y - y'}{L}\right)^2 + 1}$$

$$\approx L\left[\begin{array}{l} 1 + \frac{1}{2}\left(\frac{x - x'}{L}\right)^2 + \frac{1}{2}\left(\frac{y - y'}{L}\right)^2 - \frac{1}{8}\left(\frac{x - x'}{L}\right)^4 - \frac{1}{8}\left(\frac{y - y'}{L}\right)^4 \\ -\frac{1}{4}\left(\frac{x - x'}{L}\right)^2\left(\frac{y - y'}{L}\right)^2 + \cdots \end{array}\right]$$

$$\left(1 + x\right)^{\alpha} = 1 + \sum_{n=1}^{\infty} \frac{\alpha\left(\alpha + 1\right)\ldots\left(\alpha - n + 1\right)}{n!} x^n = 1 + \alpha x + \frac{\alpha\left(\alpha - 1\right)}{2!} x_2 + \ldots + \frac{\alpha\left(\alpha + 1\right)\ldots\left(\alpha - n + 1\right)}{n!} x^n + \ldots,\ x \in \left(-1, 1\right)$$

$$\rho=\sqrt{\left(x-x'\right)^2+\left(y-y'\right)^2+L^2}=L\sqrt{\left(\frac{x-x'}{L}\right)^2+\left(\frac{y-y'}{L}\right)^2+1}$$

$$\approx L\left[\begin{array}{l}1+\frac{1}{2}\left(\frac{x-x'}{L}\right)^2+\frac{1}{2}\left(\frac{y-y'}{L}\right)^2\cancel{-\frac{1}{8}\left(\frac{x-x'}{L}\right)^4-\frac{1}{8}\left(\frac{y-y'}{L}\right)^4}\\ \cancel{-\frac{1}{4}\left(\frac{x-x'}{L}\right)^2\left(\frac{y-y'}{L}\right)^2}+\cdots\end{array}\right]$$

$$\Rightarrow\quad e^{-ik\rho}\approx e^{-ikL\left[1+\frac{1}{2}\left(\frac{x-x'}{L}\right)^2+\frac{1}{2}\left(\frac{y-y'}{L}\right)^2\right]}=e^{-ikL}e^{-ik\left[\frac{\left(x-x'\right)^2}{2L}+\frac{\left(y-y'\right)^2}{2L}\right]}$$

$$\Rightarrow\quad K\left(x,y,x',y'\right)=\frac{i}{\lambda L}e^{-ik\rho\left(x,y,x',y'\right)}=\sqrt{\frac{i}{\lambda L}e^{-ikL}}\,e^{-ik\frac{\left(x-x'\right)^2}{2L}}\sqrt{\frac{i}{\lambda L}e^{-ikL}}\,e^{-ik\frac{\left(y-y'\right)^2}{2L}}$$

$$V(x,y)=\gamma\iint K(x,y,x',y')V(x',y')ds'$$

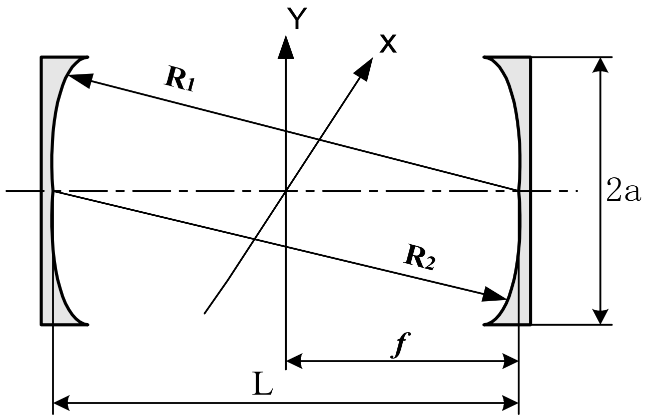
$$\left.\begin{aligned} K_x(x,x') &= \sqrt{\frac{i}{\lambda L}e^{-ikL}}\,e^{-ik\frac{(x-x')^2}{2L}} \\ K_y(y,y') &= \sqrt{\frac{i}{\lambda L}e^{-ikL}}\,e^{-ik\frac{(y-y')^2}{2L}} \end{aligned}\right\} \Rightarrow \left\{\begin{aligned} V(x) &= \gamma_x\int_{-a}^{+a}K_x(x,x')V(x')dx' \\ V(y) &= \gamma_y\int_{-b}^{+b}K_y(y,y')V(y')dy' \\ \gamma_x\gamma_y &= \gamma \end{aligned}\right.$$

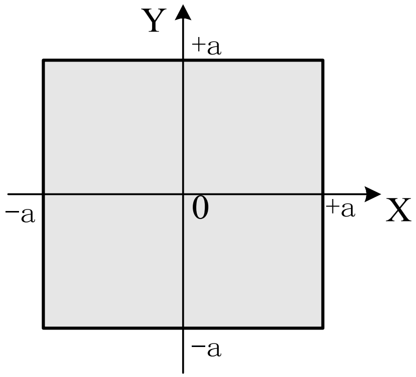
More general expression

V_m and V_n are the m^{th} and n^{th} solutions, γ_m and γ_n the corresponding complex constants:

$$\left.\begin{aligned} V_m(x) &= \gamma_m\int_{-a}^{+a}K_x(x,x')V_m(x')dx' \\ V_n(y) &= \gamma_n\int_{-b}^{+b}K_y(y,y')V_n(y')dy' \end{aligned}\right\} \Rightarrow \left\{\begin{aligned} V_{mn}(x,y) &= V_m(x)\cdot V_n(y) \\ \gamma_{mn} &= \gamma_m\gamma_n \end{aligned}\right.$$

2) Square Confocal Cavity





$$L >> a >> \lambda$$

$$\frac{a^2}{L\lambda} << \left(\frac{L}{a}\right)^2$$

Fresnel number $N=a^2/L\lambda$

Variables separation (分离变量)

$$V(x,y)=\gamma\frac{i}{\lambda L}\iint V(x',y')e^{-ik\rho(x,y,x',y')}ds'$$

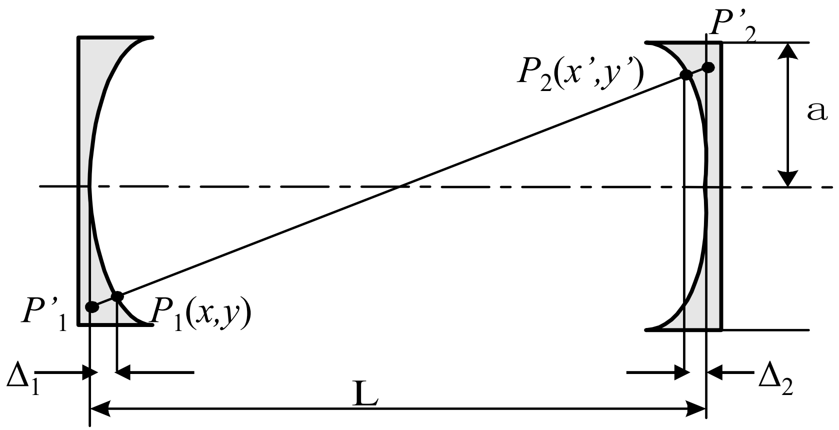
$$V(x,y)=V(x)V(y)\quad\begin{cases}V(x)=\gamma_x\int_{-a}^{+a}K_x(x,x')V(x')dx'\\V(y)=\gamma_y\int_{-a}^{+a}K_x(y,y')V(y')dy'\end{cases}$$

$$V(x,y)=\gamma\iint K(x,y,x',y')V(x',y')\,ds'$$

$$K(x,y,x',y')=\frac{i}{\lambda L}e^{ik\rho(x,y,x',y')}$$

$$\rho(x,y,x',y')=\rho_x(x,x')\rho_y(y,y')$$

$$\Rightarrow\quad K(x,y,x',y')=K(x,x')K(y,y')$$



$$\rho=\overline{p_1'}\,\overline{p_2'}-\overline{p_1'}\,\overline{p_1}-\overline{p_2'}\,\overline{p_2}$$

$$\overline{P_1'}\,\overline{P_2'}\approx L+\frac{(x-x')^2}{2L}+\frac{(y-y')^2}{2L}$$

$$\overline{P_1'}\,\overline{P_1}\approx\Delta_1\approx\frac{x^2+y^2}{2R_1}\qquad\overline{P_2'}\,\overline{P_2}\approx\Delta_2\approx\frac{x'^2+y'^2}{2R_2}$$

$$\rho(x,y,x',y')=L+\frac{1}{2L}\left[\left(1-\frac{L}{R_1}\right)(x^2+y^2)+\left(1-\frac{L}{R_2}\right)(x'^2+y'^2)-2(xx'+yy')\right]$$

$$=L+\frac{1}{2L}\left[g_1(x^2+y^2)+g_2(x'^2+y'^2)-2(xx'+yy')\right]$$

for symmetrical confocal cavity

$R_1=R_2=R=L$

$$\rho(x,y,x',y')=L-\frac{1}{L}(xx'+yy')$$

$$V(x,y)=\gamma\frac{i}{\lambda L}\iint_s V(x',y')e^{-ik\rho(x,y,x',y')}ds'$$

Square confocal cavity $\rho = L-(xx'+yy')/L$

$$V_{mn}(x,y)=\gamma_{mn}\left(\frac{i}{\lambda L}e^{-ikL}\right)\int_{-a}^{a}\int_{-a}^{a}V_{mn}(x',y')e^{ik\frac{xx'+yy'}{L}}dx'dy'$$

$c=2\pi N=2\pi\left(\frac{a^2}{L\lambda}\right)$ Fresnel number

Separate ***V****mn=Fm(X)Gn(Y)* $X=\frac{\sqrt{c}}{a}x,\quad Y=\frac{\sqrt{c}}{a}y\quad c=2\pi N\quad \gamma_{mn}=\frac{1}{\sigma_m\sigma_n}$

$$\sigma_m\sigma_n F_m(X)G_n(Y)=\frac{ie^{-ikL}}{2\pi}\int_{-\sqrt{c}}^{+\sqrt{c}}F_m(X')e^{iXX'}dX'\int_{-\sqrt{c}}^{+\sqrt{c}}G_n(Y')e^{iYY'}dY'$$

Separate into two independent equations

$$\begin{cases}\sigma_m F_m(X)=\sqrt{\dfrac{ie^{-ikL}}{2\pi}}\displaystyle\int_{-\sqrt{c}}^{+\sqrt{c}}F_m(X')e^{iXX'}dX'\\[2ex]\sigma_n F_n(Y)=\sqrt{\dfrac{ie^{-ikL}}{2\pi}}\displaystyle\int_{-\sqrt{c}}^{+\sqrt{c}}F_n(Y')e^{iYY'}dY'\end{cases}$$